MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing, Making Connections, and Normal Distributions

Oct 26 – 28 | 4.4 A Closer Look at Testing • 4.5 Making Connections • 5.1 Hypothesis Testing Using Normal Distributions | Fall 2026 · Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CLO 4, CLO 8) **2.** Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CLO 8)	e-Text: 4.4 A Closer Look at Testing Interactive Lecture: 4.4 A Closer Look at Testing Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM) Online Homework: 4.4 A Closer Look at Testing (b) (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping), and compare bootstrap and randomization distributions by their center and use. (CLO 5, CLO 8) **4.** Use a 95% confidence interval to reach a two-tailed test conclusion at α = 0.05, and decide whether a confidence interval or a hypothesis test fits a research question. (CLO 6, CLO 8)	e-Text: 4.5 Making Connections Interactive Lecture: 4.5 Making Connections Slides: 4.5 Making Connections	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples	Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
5. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution N(null value, SE). (CLO 5) **6.** Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CLO 3, CLO 8)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions Slides: 5.1 Hypothesis Tests Using Normal Distributions StatKey – Theoretical Distributions (Normal)	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 1, 11:59 PM)	WileyPLUS StatKey Blackboard
Items that cover objectives from more than one row above (objectives in parentheses)			Take-Home Quiz 6 (due Mon Nov 2, 11:59 PM) (all objectives)	Assignment Tool in Blackboard
Course resources and assessments of earlier weeks (not aligned to this week's objectives)			Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.